LAB MANUAL

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LABSHEET:03

Objectives

To understand the use of COMMIT and ROLLBACK commands in transaction control.

To use ORDER BY and GROUP BY clauses in SQL queries.

To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.

To understand and implement JOIN operations in SQL.

Commands:

• TCL commands: COMMIT, ROLLBACK, SAVEPOINT are used to control transactions.

• DCL commands: GRANT and REVOKE are used to give or remove user privileges.

• GROUP BY is used for aggregation.

• ORDER BY is used for sorting.

• JOIN is used to combine data from multiple tables.

DDL Questions (Data Definition Language)

1.Create the TEACHER table with the following attributes:

(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

CODE:

```
CREATE DATABASE LABSHEET3;
USE LABSHEET;

CREATE TABLE TEACHER(
Name VARCHAR(30),
DeptNo INT,
Date_of_Joining DATE,
DeptName VARCHAR(30),
Location VARCHAR(30),
Salary INT
);
```

2.Create the SALES table with the following attributes:

(Sales_No, Salesname, Branch, Salesamount, DOB).

CODE:

```
CREATE TABLE SALES(
Sales_No INT,
Salesname VARCHAR(30),
Salesamount INT,
Branch VARCHAR(30),
DOB date
);
```

3.**Create the DEPARTMENTS table with the following attributes:**

(dept_no, dept_name, dept_location).

CODE:

```
CREATE TABLE DEPARTMENT(
dept_no INT,
dept_name VARCHAR(30),
dept_location VARCHAR(30)
);
```

4.Create the EMPLOYEES table with the following attributes:

(emp_id, emp_name, emp_salary, dept_no).

CODE:

```
CREATE TABLE EMPLOYEES(
emp_id INT,
emp_name VARCHAR(30),
emp_salary INT,
dept_no INT
);
```

DML Questions (Data Manipulation Language)

From TEACHER Table

1.Insert five records into the TEACHER table.

```
INSERT INTO TEACHER VALUES ('Amit',101,'2015-01-10','Mathematics','Delhi',40000);
INSERT INTO TEACHER VALUES ('Riya',102,'2017-03-15','Commerce','Mumbai',35000);
INSERT INTO TEACHER VALUES ('Sohan',103,'2018-03-18','Physics','Chennai',38000);
INSERT INTO TEACHER VALUES ('Neha',104,'2016-12-12','Commerce','Kolkata',42000);
INSERT INTO TEACHER VALUES ('Krishna',105,'2019-08-16','Mathematics','Delhi',50000);
```

2.Give an increment of 25% salary to teachers working in the Mathematics department.

CODE and OUTPUT:

```
SET SQL_SAFE_UPDATES =0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
SET SQL_SAFE_UPDATES =1;
SELECT*FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Amit	101	2015-01-10	Mathematics	Delhi	50000
Riya	102	2017-03-15	Commerce	Mumbai	35000
Sohan	103	2018-03-18	Physics	Chennai	38000
Neha	104	2016-12-12	Commerce	Kolkata	42000
Krishna	105	2019-08-16	Mathematics	Delhi	62500

3.Give an increment of 15% salary to teachers working in the Commerce department.

CODE and OUTPUT:

```
SET SQL_SAFE_UPDATES =0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SET SQL_SAFE_UPDATES =1;
SELECT*FROM TEACHER;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Amit	101	2015-01-10	Mathematics	Delhi	50000
Riya	102	2017-03-15	Commerce	Mumbai	40250
Sohan	103	2018-03-18	Physics	Chennai	38000
Neha	104	2016-12-12	Commerce	Kolkata	48300
Krishna	105	2019-08-16	Mathematics	Delhi	62500

From SALES Table

1.Insert five records into the SALES table.

CODE:

```
INSERT INTO TEACHER VALUES ('Amit',101,'2015-01-10','Mathematics','Delhi',40000);
INSERT INTO TEACHER VALUES ('Riya',102,'2017-03-15','Commerce','Mumbai',35000);
INSERT INTO TEACHER VALUES ('Sohan',103,'2018-03-18','Physics','Chennai',38000);
INSERT INTO TEACHER VALUES ('Neha',104,'2016-12-12','Commerce','Kolkata',42000);
INSERT INTO TEACHER VALUES ('Krishna',105,'2019-08-16','Mathematics','Delhi',50000);
```

2.Calculate the total sales amount in each branch.

CODE and OUTPUT:

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Row

Branch	Total_Sales
Delhi	95000.00
Mumbai	125000.00
Chennai	70000.00

3.Calculate the average sales amount in each branch.

CODE and OUTPUT:

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Rows:

Branch	Average_Sales
Delhi	47500.000000
Mumbai	62500.000000
Chennai	70000.000000

4.Display the Salesname and DOB of salesmen born in the month of December with date format like 21-Dec-09.

CODE and output:

```
SELECT Salesname, DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Result Grid | Filter Rows:

Salesname	DOB
Rahul	21-Dec-09
Rohit	10-Dec-10

5.Display the Salesname and DOB of salesmen in alphabetical order of the month.

CODE:

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTH(DOB);
```

OUTPUT:

Result Grid | Filter Rows:

Salesname	DOB
Aman	2008-01-15
Neeraj	2007-03-05
Vikas	2009-08-25
Rahul	2009-12-21
Rohit	2010-12-10

From JOIN Operations

1.Display employee details and department details where the department numbers are the same in both tables.

```
SELECT E.emp_id, E.emp_name, E.emp_salary,
       D.dept_name, D.dept_location
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap C

emp_id	emp_name	emp_salary	dept_name	dept_location
101	Rohit	65000.00	Mathematics	Delhi
103	Keshav	50000.00	Mathematics	Delhi
102	Priya	45000.00	Commerce	Mumbai
105	Kanha	40000.00	Commerce	Mumbai
104	Simon	55000.00	Science	Chandigarh

2.Display the employee name and department name using a LEFT OUTER JOIN.

CODE:

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT:

Result Grid | Filter Rows:

emp_name	dept_name
Rohit	Mathematics
Priya	Commerce
Keshav	Mathematics
Simon	Science
Kanha	Commerce

3.Display the employee name and department name using a RIGHT OUTER JOIN.

CODE and OUTPUT:

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows:

emp_name	dept_name
Rohit	Mathematics
Keshav	Mathematics
Priya	Commerce
Kanha	Commerce
Simon	Science

4.Display the details of employees who draw a salary greater than the average salary.

CODE and OUTPUT:

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

Result Grid | Filter Rows: | E

emp_id	emp_name	emp_salary	dept_no
101	Rohit	65000.00	1
104	Simon	55000.00	3
NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1. Perform the ROLLBACK command on the TEACHER table.
2. Perform the COMMIT command on the TEACHER table.
3. Write a query to implement a SAVEPOINT.

CODE:

```
ROLLBACK;
COMMIT;
SAVEPOINT A;
```

DCL Questions (Data Control Language)

1.Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

CODE:

```
CREATE USER IF NOT EXISTS 'dept_user'@'localhost' IDENTIFIED BY '1234';

GRANT ALL PRIVILEGES ON LABSHEET3.EMPLOYEES TO 'dept_user'@'localhost';
```

2.Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

CODE:

```
GRANT SELECT, INSERT ON LABSHEET33.EMPLOYEES TO 'dept_user'@'localhost';
SELECT * FROM EMPLOYEES;
```

3.Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

CODE:

```
REVOKE SELECT, INSERT ON LABSHEET33.EMPLOYEES FROM 'dept_user'@'localhost';
```

4.Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

CODE:

```
REVOKE ALL PRIVILEGES ON LABSHHEET3.EMPLOYEES FROM 'dept_user'@'localhost';
SELECT * FROM EMPLOYEES;

FLUSH PRIVILEGES;
```